

PBA: 10. More on Regression

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Roadmap

1. Multiple Regression
2. Prediction Using `modelr`
3. Linear Regression and Randomized Experiments
4. Categorical Independent Variable
5. Interaction Terms in Regression
6. In-Class Exercises

Where are we? Where are we going?

1/ Multiple Regression

Beauty and Student Evaluations *Redux*

- Why do some instructors at universities receive high teaching evaluations scores from students while others receive lower ones?
 - Data contains instructor and course information on 463 courses taught at UT Austin ([ModernDive](#); see recommended resources in syllabus).

Variable	Description
<code>ID</code>	Identifier for courses taught at UT Austin
<code>score</code>	Avg. teaching score (1-5)
<code>bty_avg</code>	Instructor's avg. "beauty" score computed from a separate panel of six students (1-10).
<code>age</code>	Instructor's age

Table: UT Austin Teaching Eval. Data

- $\rightsquigarrow$ Q: Which better predicts student evaluation scores: an instructor's perceived attractiveness (beauty score) or their age?

UT Austin Student Eval. Data

Import the data:

```
1 # Don't forget to load tidyverse and ggplot2!
2 > evals <- read_csv("https://bit.ly/46rSa7s"); evals
3
4 # A tibble: 463 × 14
5   ID prof_ID score age bty_avg gender ethnicity language rank pic_outfit pic_color cls_did_eval
6   <dbl>   <dbl> <dbl> <dbl>   <dbl> <chr>   <chr>   <chr>   <chr> <chr>   <chr>   <dbl>
7 1     1     1   4.7   36     5   female minority english tenu. not formal color    24
8 2     2     1   4.1   36     5   female minority english tenu. not formal color    86
9 3     3     1   3.9   36     5   female minority english tenu. not formal color    76
10 4     4     1   4.8   36     5   female minority english tenu. not formal color    77
11 5     5     2   4.6   59     3   male   not minor. english tenu. not formal color    17
12 6     6     2   4.3   59     3   male   not minor. english tenu. not formal color    35
13 7     7     2   2.8   59     3   male   not minor. english tenu. not formal color    39
14 8     8     3   4.1   51   3.33 male   not minor. english tenu. not formal color    55
15 9     9     3   3.4   51   3.33 male   not minor. english tenu. not formal color   111
16 10    10     4   4.5   40   3.17 female not minor. english tenu. not formal color    40
17 # 453 more rows
18 # 2 more variables: cls_students <dbl>, cls_level <chr>
19 # Use `print(n = ...)` to see more rows
```

Multiple Predictors

What if we want to predict Y as a function of many variables?

$$\text{score}_i = \alpha + \beta_1 \text{bty_avg}_i + \beta_2 \text{age}_i + \varepsilon_i$$

Why?

- Better predictions (at least in-sample).
- Better interpretation as **ceteris paribus** relationships:
 - β_1 is the relationship between bty_avg and score holding age constant.
 - **Statistical control** in a cross-sectional study.

Multiple Regression in R

```
1 > eval_cnt <- evals |> # Center the variables for better interpretation of the intercept
2   mutate(bty_cnt = bty_avg - mean(bty_avg),
3         age_cnt = age - mean(age))
4 > mult.fit_cnt <- lm(score ~ bty_cnt + age_cnt, data = eval_cnt); mult.fit_cnt
5
6 Call:
7 lm(formula = score ~ bty_cnt + age_cnt, data = eval_cnt)
8
9 Coefficients:
10 (Intercept)      bty_cnt      age_cnt
11    4.174848    0.060656   -0.003059
```

- $\hat{\alpha} \approx 4.17$: average score when both the beauty score and age are at their respective mean values (i.e., 4.42 & 48.3).
- $\hat{\beta}_1 \approx .060$: average increase in score for additional beauty score an instructor received, **holding age fixed**.
 - Ceteris paribus association b/w aesthetic evaluation and score, independent of age.
- $\hat{\beta}_2 \approx -.003$: average decrease in score for one-year increase in age, **holding beauty score fixed**.

Q: Centering the variable won't change the beta coefficients, but z-score transformation would. Why?

Least Squares with Multiple Regression

- How do we estimate the coefficients?
- The same exact way as before: minimize prediction error!
- Residuals (aka prediction error) with multiple predictors:

$$Y_i - \hat{Y}_i = \text{score}_i - \hat{\alpha} - \hat{\beta}_1 \text{bty_avg}_i - \hat{\beta}_2 \text{age}_i$$

- Find the coefficients that minimize the sum of the squared residuals:

$$\text{SSR} = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta}_1 X_{i1} - \hat{\beta}_2 X_{i2})^2$$

Model Fit with Multiple Predictors

- R^2 mechanically increases when you add variables to the regression.
 - But this could be overfitting!
- Solution: penalize regression models with more variables.
 - Occam's razor: **simpler models are preferred.**
- Adjusted R^2 : lowers regular R^2 for each additional covariate.
 - If the added covariates/predictors don't help predict, adjusted R^2 goes down!

Comparing Model Fits

```
1 > library(broom)
2 > simple.bty <- lm(score ~ bty_cnt, data = eval_cnt); simple.age <- lm(score ~ age, data = eval_cnt)
3
4 > model_fits <- bind_rows(
5   simple.bty |>
6     glance() |>
7     mutate(model = "Simple (beauty only)"),
8   simple.age |>
9     glance() |>
10    mutate(model = "Simple (age only)"),
11   mult.fit.cnt |>
12     glance() |>
13     mutate(model = "Multiple (beauty + age)")
14 ) |>
15   select(model, r.squared, adj.r.squared, sigma) |>
16   mutate(across(where(is.numeric), format, digits = 6)); model_fits
17 # A tibble: 3 × 4
18   model                r.squared adj.r.squared sigma
19   <chr>                <chr>      <chr>      <chr>
20 1 Simple (beauty only) 0.0350223 0.0329290 0.534835
21 2 Simple (age only)   0.0114558 0.0093115 0.541327
22 3 Multiple (beauty + age) 0.0377810 0.0335974 0.534650
```

- Beauty explains approx. 3.5% (v.s. age $\approx$ 1.1%) of variation in scores.
- Adding age, adj. R^2 improves only slightly (0.0329 $\rightarrow$ 0.0336).
- σ (residual std. dev.) barely changes $\rightsquigarrow$ minimal improvement.

2/ Prediction Using `modelr`

Predicted Values from R

- Can we predict the evaluation score of a course currently being taught this semester (2023) with the following observed values?:

```
1 > evals_23 <- evals |> ## Suppose we have course 464 being taught in 2023
2   bind_rows(
3     data.frame(ID = 464, score = NA, age = 36, bty_avg = 4.7)
4   )
```

- The **modelr** package gives some functions that allow us to predictions in a tidy way.
 - We could plug in values into the equation, but R can do this for us.
 - Let's use **add_predictions()** to predict the 2023 results:

```
1 > library(modelr)
2 > evals_23 |>
3   filter(ID == 464) |>
4   add_predictions(mult.fit)
5
6 # A tibble: 1 × 5
7   ID score age bty_avg pred
8   <dbl> <dbl> <dbl> <dbl> <dbl>
9 1  464   NA   36     4.7  4.23
```

Predictions from Several Models

- The `gather_predictions()` will return one row for each model passed to it with the prediction for that model:

```
1 > evals_23 |>
2   filter(ID == 464) |>
3   gather_predictions(fit.bty, mult.fit)
4
5 # A tibble: 2 × 6
6   model      ID score age bty_avg pred
7   <chr>   <dbl> <dbl> <dbl>   <dbl> <dbl>
8 1 fit.bty   464    NA    36     4.7  4.19
9 2 mult.fit  464    NA    36     4.7  4.23
```

Predictions from New Data

- What about predicted values not in our data?
 - That is, what if we want to use our model to make predictions on other data?

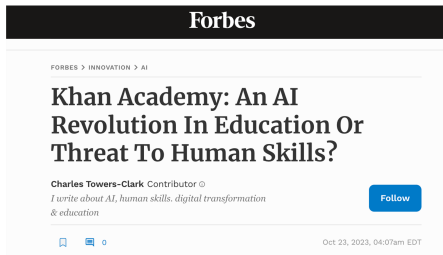
```
1 > tibble(bty_avg = c(2.5, 4.5), age = 36) |>
2   gather_predictions(fit.bty, mult.fit)
3
4 # A tibble: 4 × 4
5   model    bty_avg   age  pred
6   <chr>    <dbl> <dbl> <dbl>
7 1 fit.bty     2.5    36  4.05
8 2 fit.bty     4.5    36  4.18
9 3 mult.fit     2.5    36  4.10
10 4 mult.fit     4.5    36  4.22
```

- We can also get predicted values from the `augment()` function using the `newdata` argument:

```
1 > new_data <- tibble(bty_avg = c(2.5, 4.5), age = 36)
2 > augment(mult.fit, newdata = new_data)
3
4 # A tibble: 2 × 3
5   bty_avg   age .fitted
6   <dbl> <dbl>   <dbl>
7 1     2.5    36     4.10
8 2     4.5    36     4.22
```

3/ Linear Regression and Randomized Experiments

AI Tutoring Study Redux



<https://bit.ly/45VfES8>



Created using DALLE-3

- Can AI assistants improve students' learning outcomes?
- Experimental setting:
 - Randomly assign students to have tutoring sessions with or without AI assistants.
 - AI-assisted sessions focus on adaptive learning techniques.
- Outcome of interest:
 - Improvement in test scores or grades.

AI Tutoring Study Data (Simulated)

- Imagine that we have gone ahead and collected data from a 'second round' experiment for the AI tutoring study.

Variable Name	Description
<code>treat_ind</code>	Treatment indicator (1 for treated with AI tutoring, 0 for control)
<code>test_outcome</code>	Test outcome after being tutored (1 for Pass, 0 for Fail)
<code>education_level</code>	Tutor's education level (categorical)

```
1 > AI_data <- read_csv(url("https://bit.ly/3MuTBZY"))
2 > AI_data |> head()
3 # A tibble: 6 × 3
4   treat_ind test_outcome education_level
5     <dbl>      <dbl>      <chr>
6 1         1          0 Two-year college
7 2         1          1 Two-year college
8 3         0          0 Two-year college
9 4         0          0 Bachelor's degree
10 5         0          0 Master's degree
11 6         1          1 Bachelor's degree
```

Recall: Difference in Means Estimates

- Does AI tutoring affect test outcomes?:

```
1 > AI_data |>
2   group_by(treat_ind) |>
3   summarize(test_outcome = mean(test_outcome)) |>
4   pivot_wider(names_from = treat_ind, values_from = test_outcome) |>
5   mutate(ATE = `1` - `0`)
6
7 # A tibble: 1 × 3
8   `0`   `1`   ATE
9   <dbl> <dbl> <dbl>
10  1 0.460 0.762 0.302
```

- The average treatment effect (ATE) indicates that the AI tutoring group has a 30%p higher probability of passing the test compared to the non-AI tutoring group.
 - Assuming that the two groups are statistically identical matches (i.e., randomization ‘worked’).

Binary Independent Variables

$$Y_i = \alpha + \tau D_i + \varepsilon_i$$

- When independent variable D_i is **binary**:
 - Intercept $\hat{\alpha}$ is the average outcome in the $D = 0$ group.
 - Slope $\hat{\tau}$ is the difference-in-means of Y between $D = 1$ group and $D = 0$ group.

$$\hat{\tau} = \bar{Y}_{\text{treated}} - \bar{Y}_{\text{control}}$$

- If there are other independent variables, this becomes the difference-in-means controlling for those covariates.

Linear Regression for Experimental Data

- Under **randomization**, we can estimate the ATE with regression:

```
1 > AI_data |>
2   group_by(treat_ind) |>
3   summarize(test_outcome = mean(test_outcome)) |>
4   pivot_wider(names_from = treat_ind, values_from = test_outcome) |>
5   mutate(ATE = `1` - `0`)
6
7 # A tibble: 1 × 3
8   `0`    `1`    ATE
9   <dbl> <dbl> <dbl>
10  1 0.460 0.762 0.302
```

- This manual computation of mean differences now becomes:

```
1 > lm(test_outcome ~ treat_ind, data = AI_data) |> coef() |> round(3)
2 (Intercept)    treat_ind
3      0.460      0.302
```

4/ Categorical Independent Variable

Categorical Variables in Regression

- We often have **categorical variables**, e.g.,:
 - Education level: High school, two-year college, bachelor's, master's, PhD
 - Tutor's major: Business/economics, computer science, math, linguistics
- How to incorporate into regression: create a **series of binary variables**

Unit	Education Level	High school	Two-year col	Bachelor's	Master's	PhD
1	High school	1	0	0	0	0
2	Two-year col	0	1	0	0	0
3	Bachelor's	0	0	1	0	0
4	Master's	0	0	0	1	0
5	PhD	0	0	0	0	1
⋮	⋮	⋮	⋮	⋮	⋮	⋮

- Then include **all but one** (i.e., baseline) of these binary variables:

$$\text{outcome}_i = \alpha + \beta_1 \text{Two-year col}_i + \beta_2 \text{Bachelor}_i + \beta_3 \text{Master}_i + \beta_4 \text{PhD}_i + \varepsilon_i$$

Interpreting Categorical Variables

$$\text{outcome}_i = \alpha + \beta_1 \text{Two-year col}_i + \beta_2 \text{Bachelor}_i + \beta_3 \text{Master}_i + \beta_4 \text{PhD}_i + \varepsilon_i$$

- $\hat{\alpha}$: average outcome in the omitted group/baseline (High school).
- $\hat{\beta}$ coefficients: average difference between each group and the baseline.
 - $\hat{\beta}_1$: average difference in test outcome between those tutored by someone with two-year college degree and with high school diploma.
 - $\hat{\beta}_2$: average difference in test outcome between those tutored by someone with bachelor's degree and with high school diploma.
- Q: how do we interpret $\hat{\beta}_3$?

Regression with Categorical Variables

- Convert the categorical variable into a factor which allows us to set the baseline:

```
1 > AI_data <- AI_data |> # Specify the levels for the categorical variable
2   mutate(education_level = factor(education_level, levels = c("High school",
3                                                                "Two-year college",
4                                                                "Bachelor's degree",
5                                                                "Master's degree",
6                                                                "Doctoral/PhD")))
```

- Regress outcome on the factor variable:

```
1 > fit.cat <- lm(test_outcome ~ education_level, data = AI_data); fit.cat
2
3 Call:
4 lm(formula = test_outcome ~ education_level, data = AI_data)
5
6 Coefficients:
7             (Intercept)  education_levelTwo-year college
8                   0.57447                        0.05818
9  education_levelBachelor's degree  education_levelMaster's degree
10                   0.01528                        0.04092
11    education_levelDoctoral/PhD
12                   0.17553
```

What R Does Internally w/ Factors in `lm()`

- The factor/categorical variable, `education_level`:

```
1 > AI_data |> select(education_level) |> pull() |> head()
2 [1] Two-year college Two-year college Two-year college Bachelor's degree
3 [5] Master's degree Bachelor's degree
4 Levels: High school Two-year college Bachelor's degree Master's degree Doctoral/PhD
```

- Notice, *R* automatically created a set of indicator variables for each level of educational attainment:

```
1 > model.matrix(fit.cat) |> head()
2 (Intercept) education_levelTwo-year college education_levelBachelor's degree
3 1 1 1 0
4 2 1 1 0
5 3 1 1 0
6 4 1 0 1
7 5 1 0 0
8 6 1 0 1
9 education_levelMaster's degree education_levelDoctoral/PhD
10 1 0 0
11 2 0 0
12 3 0 0
13 4 0 0
14 5 1 0
15 6 0 0
```

Controlling Education in Estimating ATE

- We can include education-level dummies as **controls** when estimating the treatment effect:

$$Y_i = \alpha + \tau D_i + \beta_1 \text{Two-year}_i + \beta_2 \text{Bachelor}_i + \beta_3 \text{Master}_i + \varepsilon_i$$

- α : average test outcome for control group students tutored by high school educated tutors.
- τ : ATE of AI tutoring, *holding tutor education constant*.
- β_j : difference in outcomes between each education group and high school tutors, *within the control group*.

```
1 > fit.ctrl <- lm(test_outcome ~ treat_ind + education_level, data = AI_data); coef(fit.ctrl)
2
3               (Intercept)               treat_ind  education_levelTwo-year college
4               0.39172848               0.33033698               0.07463250
5  education_levelBachelor's degree  education_levelMaster's degree  education_levelDoctoral/PhD
6               0.01449457               0.15831476               0.02793454
```

- Caveat: Regression adjustment in experiments can be tricky.
 - Optional reading: [Freedman \(2008\)](#); [Lin \(2013\)](#)

5/ Interaction Terms in Regression

Within Group Comparisons for Education Level

- Q: Does AI tutoring have different effect on test outcome when tutors have different levels of education?

```
1 > AI_data |>
2   group_by(treat_ind, education_level) |> summarize(n = n()) |>
3   pivot_wider(
4     names_from = treat_ind,
5     values_from = n
6   )
7
8 # A tibble: 5 × 3
9   education_level   `0`   `1`
10  <fct>          <int> <int>
11 1 High school      63    78
12 2 Two-year college 73    74
13 3 Bachelor's degree 52    65
14 4 Master's degree  73    18
15 5 Doctoral/PhD    NA     4
16
17 > subdata_AI <- AI_data |> # account for the absence of control group counterparts
18   filter(education_level != "Doctoral/PhD")
```

↪ Omitted PhD-level tutors; our focus now is on within-group comparisons for education levels with both AI-tutored and non-AI-tutored students.

When Treatment Effects Differ Across Groups

- So far, we have estimated a single Average Treatment Effect (ATE) assuming the treatment works the same for everyone.
- To allow for this heterogeneity, we estimate the model:

$$Y_i = \alpha + \tau D_i + \sum_j \beta_j \text{Edu}_{ij} + \sum_j \delta_j (D_i \times \text{Edu}_{ij}) + \varepsilon_i$$

- Here:
 - Edu_{ij} : dummy variable for education group j (e.g., Two-year, Bachelor's, Master's; High school is baseline)
 - $\sum_j \beta_j \text{Edu}_{ij}$: captures baseline outcome differences across education groups
 - τ : treatment effect for the **baseline** group (High school tutors)
 - δ_j : **additional effect of treatment** for education group j compared to the baseline

⇒ This specification lets us estimate group-specific, Conditional Average Treatment Effects (CATEs).

Conditional ATE

```
1 > fit.cate <- lm(test_outcome ~ treat_ind * education_level,  
2 data = subdata_AI); coef(fit.cate)  
3  
4 (Intercept) treat_ind  
5 0.49206349 0.14896215  
6 education_levelTwo-year college education_levelBachelor's degree  
7 -0.04000870 -0.14590965  
8 education_levelMaster's degree treat_ind:education_levelTwo-year college  
9 0.02848445 0.20979387  
10 treat_ind:education_levelBachelor's degree treat_ind:education_levelMaster's degree  
11 0.28949939 0.33048991
```

- **Q:** Based on the regression output above, what is the estimated Conditional Average Treatment Effect (CATE) for students tutored by someone with Master's degree?

$$\widehat{\text{CATE}}_{\text{Master's}} = \hat{\tau} + \hat{\delta}_{\text{Master's}}$$

6/ In-Class Exercises

In-Class Exercises: Background

- You are working for an investor who is thinking of building a new condominium in NYC and want to predict the market value before making any decision.
 - The new condominium will be located in **Manhattan** and will be **220,000 square feet** large with **300 units**.
- The NYC housing dataset contains some information about the condominiums located in New York city and thus can be used to make the prediction. Some relevant variables:
 - `Market.Value.per.SqFt`: total market value per square foot of the land and building
 - `Boro`: the location (borough) of the condominium in NYC (e.g., Manhattan)
 - `Total.Units`: total number of units in the building
 - `Gross.SqFt`: the size of the building in square foot

In-Class Exercises: Tasks

1. Import the data using `read_csv()` from `{ "https://bit.ly/49QWVuc" }` and store the resulting tibble as `housing`. Using `select()` and `rename()`, keep 'Market.Value.per.SqFt', 'Boro', 'Total.Units', and 'Gross.SqFt' renamed as 'value', 'boro', 'units', and 'SqFt', respectively, for the subsequent analysis. Don't forget to overwrite `housing` with the subsetted data.
2. Specify two regression models where you consider 1) only the location (i.e., boro) as the predictors and 2) the location as well as the size (i.e., SqFt) & units (i.e., units) as the predictors. Write out the equations for the two models.
3. Estimate the models using `lm()` and interpret the coefficients.
4. Compare the 'fit' of the two models by looking at the adjusted R^2 with `glance()`. That is, do size and number of units 'add value' to our prediction?
5. Predict the market value of the condominium using `modelr` functions. What is the value of the condominium predicted to be given the specifications detailed in the previous slide?
6. The investor wants to build only if the valuation is predicted to be higher than 140 per Sq Ft. Should you recommend him to invest or not?

Enjoy Your Weekends! :)

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Appendix

Presenting Regression Results with stargazer

	Dependent variable:
	test_outcome
treat_ind	0.149* (0.078)
education_levelTwo-year college	-0.040 (0.079)
education_levelBachelor's degree	-0.146* (0.087)
education_levelMaster's degree	0.028 (0.079)
treat_ind:education_levelTwo-year college	0.210* (0.109)
treat_ind:education_levelBachelor's degree	0.289** (0.116)
treat_ind:education_levelMaster's degree	0.330** (0.145)
Constant	0.492*** (0.058)
Observations	496
R ²	0.123
Adjusted R ²	0.111
Residual Std. Error	0.462 (df = 488)
F Statistic	9.814*** (df = 7; 488)
Note:	*p<0.1; **p<0.05; ***p<0.01

- Use `stargazer()` to produce well formatted results table (type = "html" for html output):

```
# Don't forget to install `stargazer`  
> stargazer(fit.cate, type = "text")
```

- The coefficient for `treat_ind`: the baseline ATE for tutors w/ high school graduates.
- The coefficients for the interaction terms (purple box): additional effect of treatment for each education level compared to high school.
- Common rule of thumb: coefficient estimate is stat. sig. (at 5%) if it is larger than twice the standard error in parenthesis.
 - Est. coef. for `treat × master's` = $0.289 > (0.116 \times 2)$

Visualizing Estimated Coefficients w/ jtools

- We can use `plot_summ()` function to visualize the estimated regression coefficients along with their standard errors:

```
1 # Install `jtools` and `broom.mixed` packages
2 > library(jtools)
3 > plot_summs(fit.cate)
```

